$(u,v),\ (w,x)$ critical edges of G^*

G^*

G^+

$\varphi^+ \to S(G^+)$

$u < w = \text{true}$

$v < x = \text{true}$

$uv \iff wx$

$uv \iff wx$

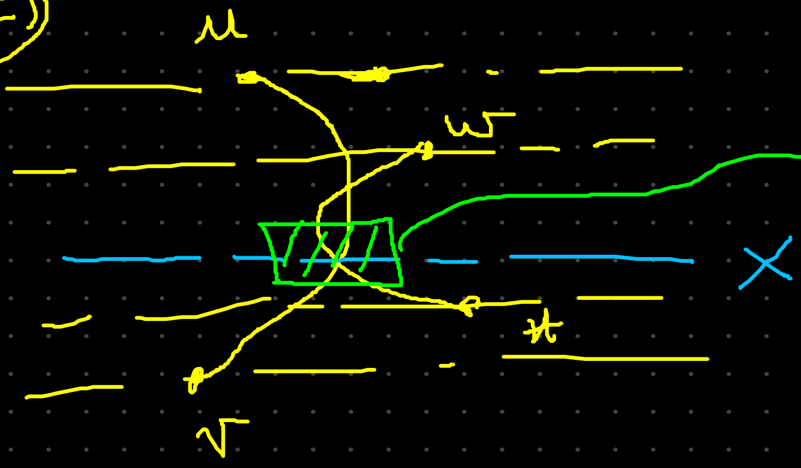
$$\varphi^+(u'\ w') = \varphi^+(v'\ x')$$

what to prove:

$(u,v),(w,x)$ cross even times

$\iff \varphi^+(u'\ w') = \varphi^+(v'\ x')$

$\Rightarrow$

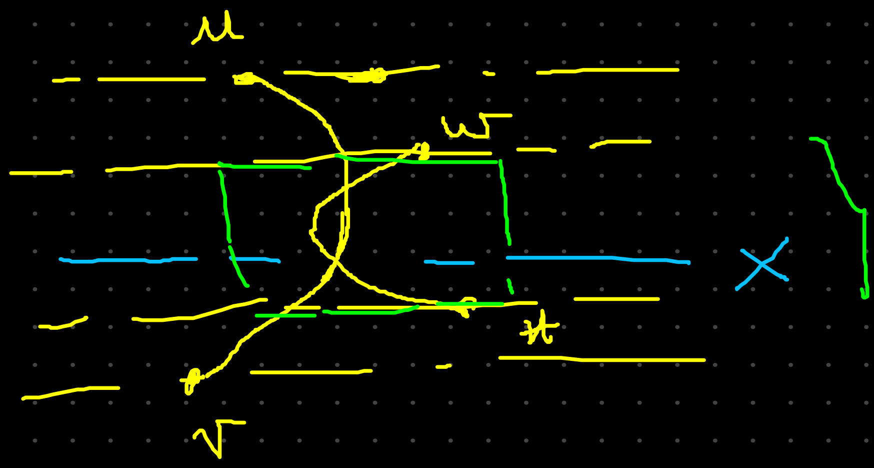


$\Downarrow$

$$\varphi^+(u_1,w_1) \neq \varphi^+(u_2,w_2)$$

if i understand , we proved it for one crossing, and we can generalize it, each 2 crossing will "nullify" each other, So having even number of crossing will give us the equality of the truth assignment. And the corssing can only be in the limit area,

$\Leftarrow)$ $\varphi^+(u'w') = \varphi^+(v'x')$



The crossing will occur on the limit zone of the critical edges

if $\varphi^+(u'w') = \varphi^+(v'x')$

it means that there are even crossings in the critical area and also between (uw) and (vx)

— Lemma 3 —

G proper level graph & Γ^* H-T drawing of G^*

$\Rightarrow$ $S(G)$ is satisf

Γ^+ drawing of G^+ induced by Γ^*

ψ^+ truth assi. induced by Γ^+

φ^+ satisfies $S(G^+)$

ψ^+ not necessary crossing contraint

$u'', w'' \in V(h^+)$ with $l(u'') = l(w'')$

$u'' \approx w'' \in V(h^*)$

$\varphi^+(u'', w'') = \psi^+(u'', w'')$

$u'', w'' \notin V(h^*)$

subdivision of $u, w \in V(h^*)$

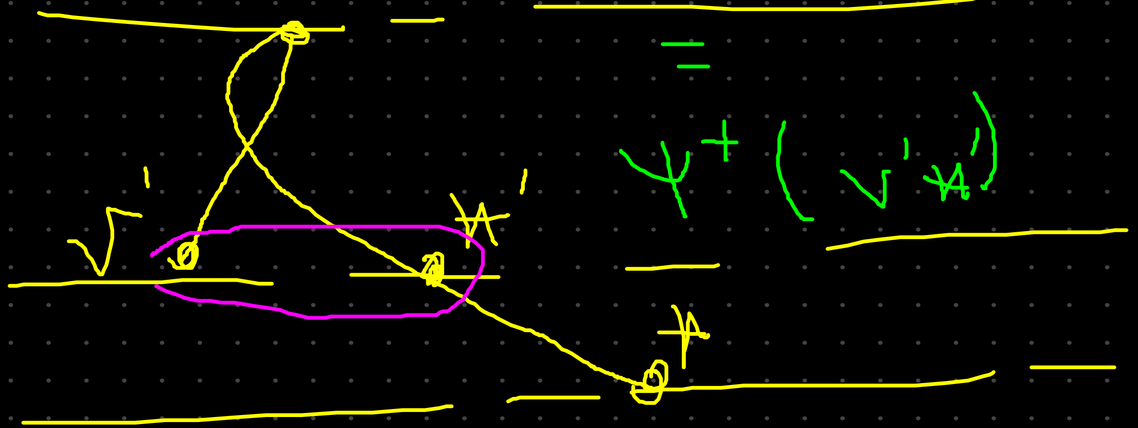
h^+: $l(u'') = l(w'')$

$\varphi^+(u' w'') = \psi^+(u' w')$

$u'' = w''$ $\varphi^+(u'' v'')$

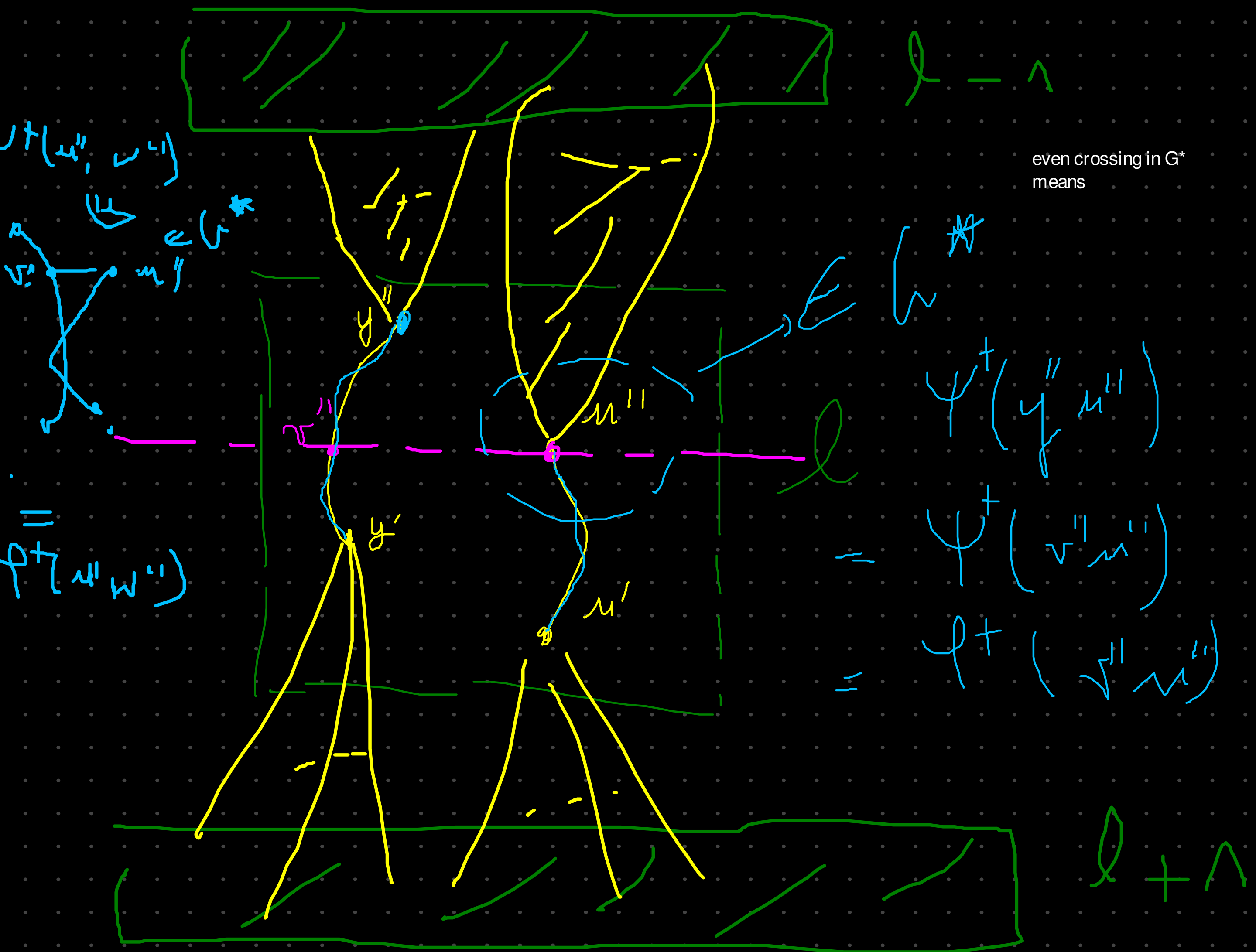
$=$

$\psi^+(v' x')$



$$\varphi^+(u'' w'') = \psi^+(u'' w'')$$

$\psi^+(u'', w'') \uplus \in G^*$



$= \varphi^+(u'' w'')$

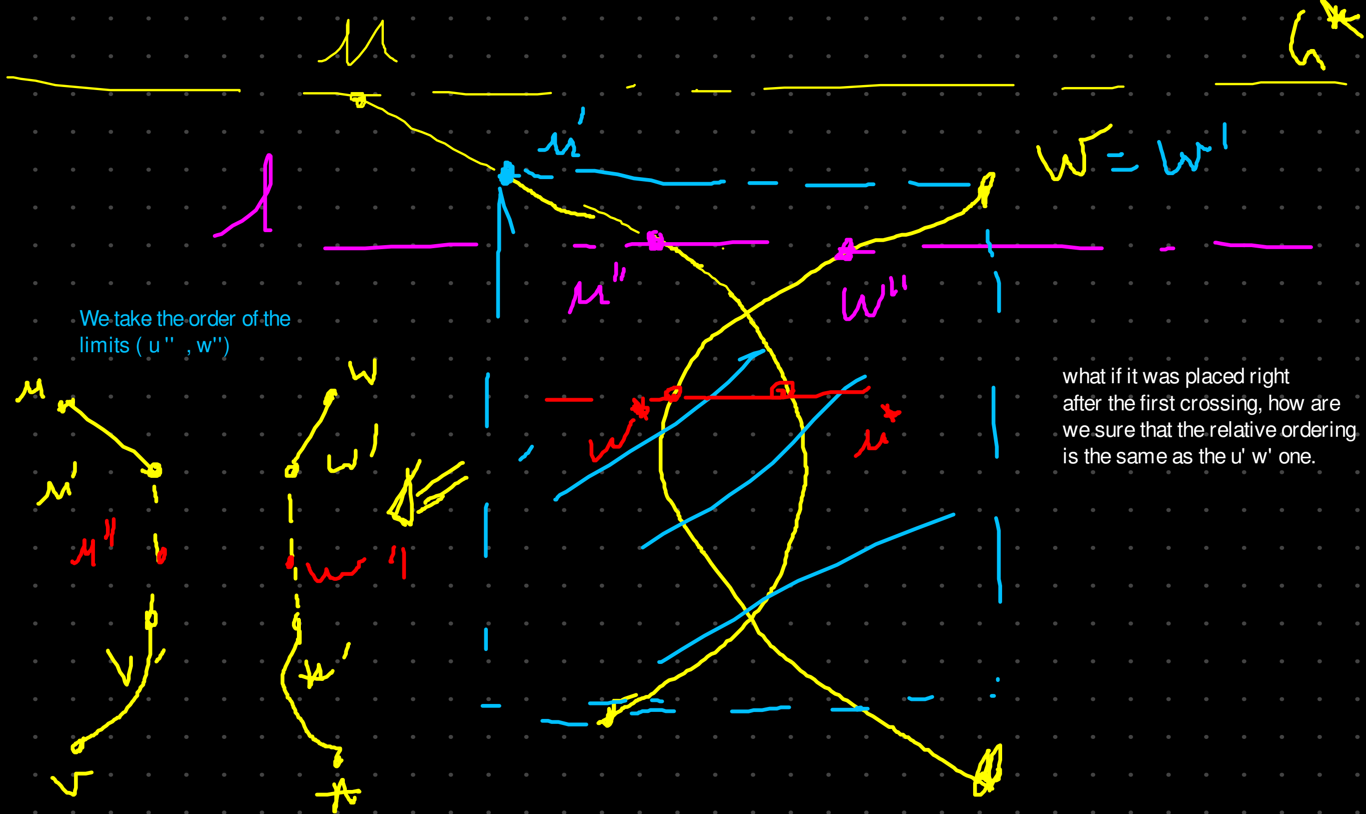
even crossing in G*
means

$\in G^*$

$\psi^+(y'' u'')$

$= \psi^+(v'' u'')$

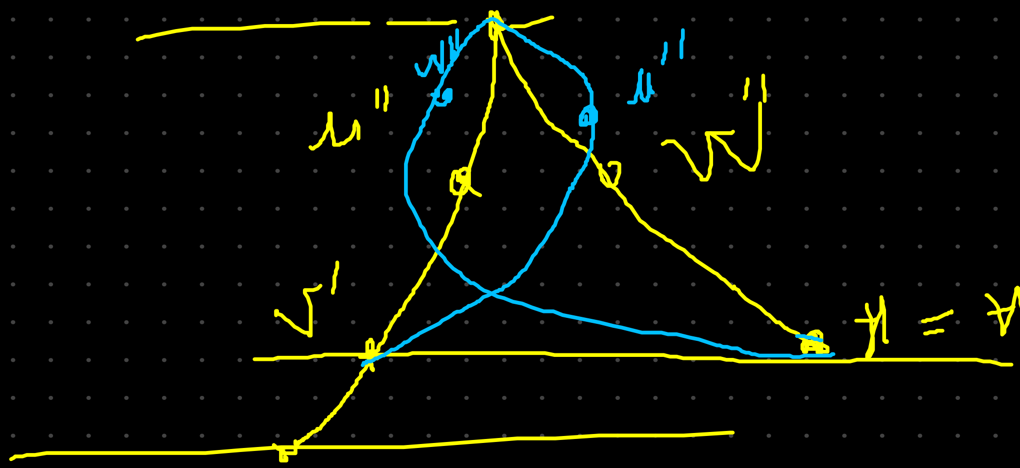
$= \varphi^+(v'' u'')$



We take the order of the limits (u'' , w'')

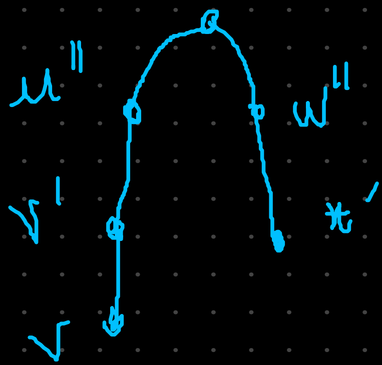
what if it was placed right
after the first crossing, how are
we sure that the relative ordering
is the same as the u' w' one.

$$\varphi^+(u'' \, w'') = \psi^+(v' \, x')$$



||

φ^+ $\quad u = w$



made

$u \; v \; w \in V(x^*)$

$\varphi^+(u'' \; w'') =$

$\psi^+(u'', w'')$

$u \neq v$

$\varphi^+(u'', w'')$

$= \psi^+(u', w')$

$u = v$

$\varphi^+(u'', w')$

$= \psi(v', x')$

$\forall$ critical edge $(u'', v''), (w'', x'') \in E(h^+)$

that $\varphi^+(u''w'') = \varphi^+(v''x'')$

$$\varphi^+(u''w'') = \Psi^+(u''w'') = \Psi^+(v''x'') = \varphi^+(v''x'')$$

if $u'' \cap w'' \in V(h^*)$ and $v'' \cap x'' \in V(h^*)$

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if $u'', w'' \notin V(h^*)$

$$\varphi^+(u''w'') = \Psi^+(u'w') = \Psi^+(v''x'')$$

if $u'', w'', v'', x'' \notin V(h^*)$

$$\left\{\begin{array}{l} \varphi^+(u''w'') = \Psi^+(u'w') \\ \varphi^+(v''x'') = \Psi^+(v'x') \end{array}\right. \quad \Psi^+(u'w') = \Psi^+(v'x')$$

Lemma 2